

MANTHAN SHRIRAM JOSHI

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Education

Northeastern University, Boston | MA | Master of Science in Cell and Gene Therapies

December 2026

Coursework: Stem Cells and Regeneration, Cell and Gene Therapies, Immunology, Regulatory Landscapes, Advanced Drug Delivery Systems, Bioinformatics, Scientific Communication and Biotech Enterprise

Harvard Business School, Boston | MA | Certificate in Management Essentials

August 2025

Manipal Institute of Technology (MAHE), Manipal | India | Bachelor of Technology in Biotechnology | Minor:

July 2023

Pharmaceutical Biotechnology

Four-year accredited engineering program. Coursework: Bioprocess Engineering, Downstream Processing, Bioreaction Engineering, Biochemistry, Genetic Engineering, Microbiology, Bioinformatics

Skills

Process & Manufacturing	Aseptic technique; process development, optimization, and yield improvement; assay development (dose-response, cytotoxicity, kill curve design); protocol and SOP authorship; experimental troubleshooting and root-cause investigation; cell expansion, banking, and cryopreservation; scientific documentation and technical reporting.
Cell Culture & Genome Engineering	Aseptic handling of mammalian cell lines (CHO) across 5+ passages at >80% viability; end-to-end CRISPR/Cas9 knock-in and knock-out workflow design; Cas9-RNP assembly with target-specific sgRNAs; CRISPRmax lipofection; drug-selection optimization using CellTiter-Glo 2.0 cytotoxicity assays.
Molecular Biology & Analytical	Bacterial transformation, colony screening, miniprep purification; restriction-enzyme digestion for construct verification; genomic DNA extraction; Phusion PCR; primer design in Benchling; Sanger sequencing and ICE indel analysis; agarose gel electrophoresis; Nanodrop quantification; recombinant protein expression and Ni-NTA purification (IMAC); SDS-PAGE; Western blotting; ELISA.
CGT Domain Knowledge	CAR-T (autologous and allogeneic platforms); iPSC differentiation; CRISPR/Cas9 genome engineering; lentiviral vector handling and titer determination; advanced drug delivery systems; cGMP and CGT regulatory frameworks (coursework).
Commercial & Business	Market research and competitive landscaping; investor deck development; grant proposal support; technical writing for non-specialist audiences; client and stakeholder communication; cross-functional project coordination.
Instrumentation & Software	Bioreactor/fermenter operation; HPLC; gas chromatography; fluorescence microscopy; flow cytometry sample preparation; FlowJo (data analysis); Benchling (primer design, construct mapping); GraphPad Prism (dose-response curve fitting); Microsoft Excel; Microsoft Office; ICE web tool; R (basic).

Professional Experience

Northeastern University, Teaching Assistant, Cell and Gene Therapies, Boston, MA

Jan 2026 – Apr 2026

- Taught graduate-level CGT concepts in office hours and discussion sessions, translating current CAR-T and iPSC-to-T-cell literature for students from mixed scientific backgrounds.
- Reviewed student project deliverables and delivered structured written feedback; prepared literature summaries on allogeneic and autologous CAR-T platforms to support classroom case discussions.

Northeastern University, Graduate Research Project, Boston, MA

Sept 2025 – Dec 2025

- Maintained CHO cell cultures across 5+ passages at >80% viability; assembled and delivered Cas9-RNP complexes with 2 HPRT1-targeting sgRNAs via CRISPRmax lipofection for an end-to-end HPRT1 knockout and knock-in workflow.
- Developed and optimized a 6-thioguanine selection protocol using CellTiter-Glo 2.0 dose-response across 8 serial 3-fold dilutions, graphed in GraphPad Prism, selecting 100 μ M (~7% viability) to enrich HPRT1-KO cells; confirmed by fluorescence microscopy and prepared single-cell suspensions for flow cytometry.
- Extracted genomic DNA from selected pellets, amplified HPRT1 by Phusion PCR, and submitted 8 amplicons for Sanger sequencing and ICE indel analysis to confirm gene disruption.
- Validated 5 plasmid constructs by restriction digest and gel electrophoresis; designed 2 primer sets in Benchling for diagnostic and knock-in confirmation amplicons.

Reagene Biosciences Pvt Ltd, Associate Scientist, Bangalore, India

Aug 2023 – Aug 2026

- Developed, optimized, and validated ELISA platforms including COVID-19 diagnostics, and designed cell-based bioassays to evaluate therapeutic efficacy and cytotoxicity.
- Ran in-vitro ADME studies to characterize compound pharmacokinetics, using HPLC for quantitative and qualitative analysis of biochemical compounds and fluorimetric glucose uptake and inhibition assays.
- Operated and optimized 3D bioprinting to build in-vitro tissue models and organoids for disease modeling, owning experimental design, protocol optimization, and quality control documentation.
- Maintained and cryopreserved mammalian cell lines at high viability to supply the bioink formulation process, and authored SOPs for routine laboratory procedures.
- Built market research and competitive analysis for the tissue platform, and contributed to investor decks and grant proposals used in fundraising.

Tranalab Pvt Ltd, Center for Human Genetics, Bachelor's Thesis, Bangalore, India

Jan 2023 – Aug 2023

- Designed and executed a complete expression and purification process for M4, a 35 kDa recombinant membrane protein domain, as a bachelor's thesis project ranked first in the cohort.
- Identified solubilization as the yield-limiting step and engineered a modified buffer using non-ionic detergents, increasing soluble protein recovery by roughly 48% (A280 0.46 to 0.68).
- Ran the upstream and downstream train end to end: expression in *E. coli* BL21(DE3) under IPTG induction, cell lysis by probe sonication, inclusion body solubilization, and Ni-NTA affinity chromatography (IMAC) with a stepwise urea wash gradient and imidazole elution.
- Confirmed product identity by SDS-PAGE with Coomassie staining, quantified yield by A280, buffer-exchanged by overnight dialysis, and assessed refolding by sugar-affinity chromatography to verify retained binding function.

Leadership and Projects

Vice President, Northeastern University, Graduate Biotech-Bioinfo Association, Boston, MA

Dec 2024 – Jan 2026

- Led executive board operations and organizational strategy, managing communications and cross-committee coordination while planning networking events, workshops, and industry panel discussions for member engagement.

CurcuNano, Chief Executive Officer, BIOT 5219 Biotech Enterprise, Northeastern University

Spring 2026

- Led the venture team as CEO for a curcumin nanoformulation concept, developing the business case and market positioning across a semester-long project.
- Owned the funding ask in the final investor pitch, covering raise amount, use of funds, and investment rationale, defended to a faculty panel.